

From: Elaine J. Venema <evenema@fveng.com>
Sent: 1/16/2025 4:08:24 PM
To: "Lowe, Claire (EGLE)" <LoweC9@michigan.gov>
Cc: "utilities@cadillac-MI.net" <utilities@cadillac-mi.net>; "Patrick Miller" <pmiller@cadillac-mi.net>; "Cindy Tomaszewski" <ctomaszewski@Cadillac-MI.net>
Subject: RE: Cadillac MAHL - December 2024
Attachments: Cadillac MAHL EGLE Comments Response Jan2025.pdf

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Hi Claire,

Thanks for the comments. The attached letter includes our responses. If these responses adequately address your comments, we will revise the report accordingly and the City will resubmit in MiEnviro.

If you'd like to discuss any of these items in more detail, we can set up a virtual meeting in Teams or Zoom.

Thanks!

Elaine

Elaine J. Venema, PE

Sr. Project Manager, Associate

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O: 616.977.1000

C: 616.446.9669

D: 616.588.1913

2960 Lucerne Drive SE, Grand Rapids, MI 49546

FVCPM
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www.fveng.com



From: Lowe, Claire (EGLE) <LoweC9@michigan.gov>

Sent: Friday, January 10, 2025 10:47 AM

To: Elaine J. Venema <evenema@fveng.com>; utilities@cadillac-MI.net; Patrick Miller <pmiller@cadillac-mi.net>

Subject: Cadillac MAHL - December 2024

Good Morning,

I completed my review of the MAHL Report submitted in December. I believe many of the proposed limits are approvable. I have comments on the proposed limits for BOD, phosphorus, silver and phenol which are attached. I think most of these are easily resolved, with the possible exception of the phenol limit. Please let me know if you would like to discuss further. I should also get back to you on the manual review soon.

Thank you,

Claire Lowe

Environmental Quality Analyst

Water Resources Division, Gaylord District Office

Michigan Department of Environment, Great Lakes, and Energy

[989-330-4639](tel:989-330-4639) | LoweC9@Michigan.gov

Michigan.gov/EGLE



MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY



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ATTACHMENT NAME:

Cadillac MAHL EGLE Comments Response Jan2025.pdf

ATTACHMENT TYPE:

Adobe Portable Document Format (PDF) compound image



January 16, 2025

Via email: LoweC9@michigan.gov

Ms. Claire Lowe
Michigan Department of Environment, Great Lakes, and Energy
Water Resources Division
Gaylord District Office
2100 West M-32
Gaylord, MI 49735

**RE: City of Cadillac WWTP (NPDES Permit No. MI0020257)
MAHL Local Limits Evaluation Response to EGLE Comments**

Dear Ms. Lowe:

Thank you for providing feedback on the Maximum Allowable Headworks Loading (MAHL) and local limits evaluation for Cadillac. We offer the following responses to the comments provided via email on January 10, 2025.

1. Industrial users

Page 3. Several SIUs were recategorized to non-SIUs. Please ensure that further explanation on why these were recategorized is included in the IPP annual report e.g. categorical discharge ceased, was incorrectly identified as a categorical process, flow decreased below SIU classification etc.

The City will provide additional information in the annual IPP report; however, a brief summary is provided for each IU classification change:

Avon was originally classified as an SIU based on its potential to discharge cyanide containing wastewater. Cyanide gas is used to test the gas mask seals in a fume hood. The gas is collected in water, then batch treated. The batch treated wastewater used to be tested, then discharged to the City sewer. Avon has recently changed its SOP and is no longer discharging treated cyanide wastewater to the City. Avon is collecting the wastewater and having it hauled offsite for disposal.

Munson Healthcare was identified as a categorical discharged based on a misunderstanding of the 40 CFR Part 460. Healthcare facilities must meet categorical standards if they are direct discharges; not indirect discharges to a POTW. Onsite review of the wastewater generated onsite did not indicate any reason to classify Munson as either an SIU or a CIU. Wastewater is primarily domestic / sanitary wastewater with small amounts of RO reject, softener backwash, and boiler blowdown.

603 Bay Street, First Floor
Traverse City, MI 49684
P: 231.932.8600
F: 231.932.8700
www.fveng.com

Piranha Hose was incorrectly identified as being re-classified as non-significant in the MAHL report. This will be corrected in the final version of the report, and the City will continue to regulate them through the IPP.

Spencer Plastics does injection molding of plastic parts with no categorical processes or products. The majority of the wastewater is sanitary in nature with small amounts of non-contact cooling water. A review of historical analytical data and SDS sheets did not indicate any potential for interference or pass-through to the POTW.

Rec Boats Holding (Trailer Division) was classified as a metal finisher under 40 CFR Part 433 due to a metal etching process prior to painting. The etching and paint process has since been removed from the facility. The facility currently only does boat trailer assembly with no categorical or significant processes.

2. Phenols

Page 5, Table 2. The parameters identified include total phenols. The current approved local limit and WQBEL provided for this study are for phenol. EGLE does not recommend developing local limits for total phenols/total phenolic compounds as this is not a meaningful grouping in terms of toxicity or the behavior of the various pollutants. It is not clear based on the information provided whether samples collected out in the collection system for background concentration or within the WWTP to determine removal of pollutants were analyzed for phenol or for total phenolic compounds. Please clarify if analytical results can be/are provided for phenol alone. If results can't be provided for phenol it may not be possible to revise the phenol limit on the basis of the information currently available.

The analytical results presented in the MAHL report are for total phenolic compounds, using EPA Method 420.4. An excerpt of the method is below.

2.0 SUMMARY OF METHOD

2.1 This semi-automated method is based on the distillation of phenol and subsequent reaction of the distillate with alkaline ferricyanide and 4-aminoantipyrine to form a red complex which is measured at 505 or 520 nm.

2.2 Color response of phenolic materials with 4-aminoantipyrine is not the same for all compounds. Because phenolic type wastes usually contain a variety of phenols, it is not possible to duplicate a mixture of phenols to be used as a standard. For this reason, phenol has been selected as a standard and any color produced by the reaction of other phenolic compounds is reported as phenol. This value will represent the minimum concentration of phenolic compounds present in the sample.

Based on this, the concentration of phenol in a sample will be less or equal to the total phenolics result.

In addition, we propose to remove phenol from the local limits because there is no potential for any of the industrial users to discharge phenol or phenol-containing compounds. A review of the SDS sheets and all chemicals used at all the permitted industrial users did not reveal any phenol or phenolic chemicals.

3. Silver

Page 13, Table 13. The MAIL after safety factor stated for Silver is given as 0.070 lbs/day based on the WQBEL for Chronic toxicity. This agrees with the calculations given in Table A6. If I complete this calculation, I make this result 0.0485 lbs/d. I don't know what the source of the discrepancy is for this. However, based on the proposed local limit for silver of 3 ug/L the limit is protective at both the above stated MAILs.

We re-calculated the silver chronic toxicity AHL. The monthly average WQBEL provided by EGLE is 1.4 µg/L, or 0.0014 mg/L. The literature removal of 75% was utilized in the calculation, as WWTP influent and effluent silver concentrations were below detection.

Inserting these values into the NPDES formula:

$$8.34 * 0.0014 \text{ mg/L} * 1.72 \text{ MGD} / (1-0.75) = 0.0803 \text{ lb/d}$$

$$\text{Background loading} = 8.34 * 0.0004 \text{ mg/L} * 0.8362 \text{ MGD} = 0.0027 \text{ lb/d}$$

$$\text{MAIL (with 10\% SF)} = (0.0803 - 0.0027) * 0.9 = 0.0698 \text{ lb/d, rounded to 0.07 lb/d}$$

The calculated local limit is 0.01 mg/L, or 10 µg/L.

The City wishes to keep the current local limit of 3 µg/L for silver.

4. BOD

Page 15, Table 15. The % removal stated for BOD for June to September is 97.7%. This appears to be incorrect based on the MOR data provided with the first version of the MAHL study. The removal across the entire WWTP for June to September is stated as 97.6%, so this appears to have been incorrectly entered into the calculations. When the removal is adjusted to 97.6 the MAHL is reduced from 4366 lb/d to 4184 lbs/d and June to September. The resulting MAIL and local limits are reduced from 2710 to 2528 lbs/d and 400 mg/L to 353.7 mg/L, respectively.

This has been corrected, using 97.6% removal, resulting in the values calculated by EGLE. The calculated local limit is 354 mg/L, and the proposed BOD local limit is 350 mg/L.

5. Phosphorus

Page 6, Table 3. The table states the flow weighted average background samples for phosphorus is 6.3 mg/L. Based on the flow data provided on page 4, and background concentrations provided in appendix A, it appears the phosphorus flow-weighted average concentration should be 6.8 mg/L.

The discrepancy between our flow-weighted average calculations for background phosphorus appears to be due to rounding / significant figures. Background sample location #1 average was 14.6 mg/L; background sample location #2 average was 5.5 mg/L. Applying the flow rate for each location results in a flow-weighted average of 6.34 mg/L, rounded to 6.3 mg/L.

The full background sampling results are attached for reference.

Page 17, Table 18 states that the phosphorus % removal for the period of May to March is 94.7%. The MOR data provided in the first version of the MAHL study does not provide an average for this period, but when calculated out based on the raw data this appears to be 94.5% removal. The resulting MAHL based on this removal is reduced from 135.3 lbs/d to 130.4 lbs/d. Once the correction is made to the flow weighted average background concentration from 6.3 to 6.8 mg/L the background loading increases from 45 lbs/d to 49 lbs/d. The MAIL is therefore reduced from the 83 lbs/d, as stated in Section 5.5, Table 19, to 74.9 lbs/d and the resulting local limit once the safety factor is included is reduced from 11 mg/L to 10.5 mg/L.

The 94.7% removal from October to November was incorrectly used. The average removal from May to March is 94.5%. Inserting this value into the NPDES formula results in 130 lb/d phosphorus MAHL. After subtracting the background loading of 46 lb/d and applying the 5% safety factor, the MAIL is calculated as 78 lb/d, resulting in a proposed local limit of 11 mg/L.

Please call or email me with any questions regarding these responses. If there are no additional questions or comments, we will finalize the report based on these responses and the City will resubmit it in MiEnviro.

Sincerely,

FLEIS & VANDENBRINK



Elaine J. Venema, P.E.
Project Manager

Enclosure: Background Sampling Results

Cc w Encl: Jeff Dietlin, City of Cadillac
Patrick Miller, City of Cadillac
Cindy Tomaszewski, City of Cadillac

Domestic/Background Samples

		Grab Sample Parameters		Metals (Total)										Compatible Pollutants				Organic Pollutants									
Collection		Cyanide, Total	Cyanide, Available	Mercury	Arsenic	Cadmium	Chromium	Copper	Lead	Molybdenum	Nickel	Selenium	Silver	Zinc	NH ₃ -N	BOD ₅	TSS	Phosphorus	PCB	Chloroform	1,2-Dichloro benzene	Ethylbenzene	Styrene	Toluene	Xylene	Phenolics	Lindane
Date	Location	mg/L	µg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L	mg/L
10/5/2022	Domestic #1	0.0052	16.8	0.00045	0.0054	0.0016	0.0151	0.694	<0.001	0.0055	0.0192	<0.005	0.0038	1.37	47.4	620	4920	24.5	<0.0001	<0.0048	<0.005	<0.005	<0.005	0.0067	<0.010	0.0416	<0.00051
10/5/2022	Domestic #2	<0.0050	4.6	<0.0002	0.0013	<0.0002	<0.002	0.0962	0.0013	0.0012	0.0027	<0.001	<0.0005	0.125	38.8	186	400	7.0	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.143	<0.00051
10/13/2022	Domestic #1	<0.0050	3.4	0.00074	0.0043	0.00052	0.0158	0.0497	0.13	0.0011	0.0131	<0.001	<0.0005	0.239	10.1	141	414	4.7	<0.000096	<0.0048	<0.005	<0.005	<0.005	0.0054	<0.010	<0.0667	<0.000048
10/13/2022	Domestic #2	<0.0050	4.3	<0.0002	0.0016	<0.00020	<0.002	0.0892	0.001	0.0014	0.0024	<0.001	<0.0005	0.141	34.6	254	216	6.3	<0.000096	<0.0048	<0.005	<0.005	<0.005	0.0054	<0.010	0.149	<0.000048
10/14/2022	Domestic #1	<0.0050	3.5	0.0002	<0.002	0.00049	0.0062	0.255	0.0089	0.0025	0.0064	<0.002	0.0023	0.608	41	410	1280	14.7	<0.00011	<0.0048	<0.005	<0.005	<0.005	0.0066	<0.010	<0.0667	<0.000056
10/14/2022	Domestic #2	<0.0050	4.1	<0.0002	0.0013	<0.00020	<0.002	0.0638	<0.001	<0.001	0.0019	<0.001	<0.0005	0.0809	29.5	116	176	4.4	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.000050
10/21/2022	Domestic #1	<0.0050	5.3	0.00037	0.0016	0.0005	0.0055	0.249	0.0124	0.0025	0.0064	0.0011	0.0013	0.568	35.9	440	1320	13.2	<0.000098	<0.0048	<0.005	<0.005	<0.005	0.0056	<0.010	<0.0667	<0.000049
10/21/2022	Domestic #2	<0.0050	4.5	<0.0002	0.001	<0.0002	<0.002	0.0681	<0.001	<0.001	0.002	0.0011	<0.0005	0.0897	25.8	138	184	4.8	<0.0001	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.200	<0.000050
10/22/2022	Domestic #1	<0.0050	3.2	<0.0002	<0.001	0.00025	<0.002	0.078	0.0021	0.0017	0.0055	<0.001	<0.0005	0.268	37.8	>750	440	17.6	<0.000098	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.0683	<0.000049
10/22/2022	Domestic #2	<0.0050	4.6	<0.0002	<0.001	<0.0002	<0.002	0.0496	<0.001	<0.001	0.002	<0.001	<0.0005	0.0856	30.2	353	75	5.4	<0.00011	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.0667	<0.000053
10/25/2022	Domestic #1	<0.0050	4.5	<0.0002	0.0018	0.00052	0.0056	0.263	0.0123	0.0025	0.0056	0.0014	0.0026	0.584	32.5	530	1220	13	<0.000098	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	<0.100	<0.000049
10/25/2022	Domestic #2	<0.0050	5.9	<0.0002	<0.001	<0.0002	<0.002	0.0504	<0.001	<0.001	0.0014	<0.001	<0.0005	0.0623	39.2	163	141	5.2	<0.000099	<0.0048	<0.005	<0.005	<0.005	<0.005	<0.010	0.239	<0.000050
Average - DS 1		0.0030	0.006	0.00034	0.0024	0.0006	0.0082	0.2648	0.0278	0.0026	0.0094	0.0012	0.00175	0.606	34	481.8	1599	14.6	0.00020	0.0024	0.0025	0.0025	0.0025	0.0049	0.005	0.043	0.00006
Average - DS 2		0.0025	0.005	0.00010	0.0010	0.0001	0.0010	0.070	0.0007	0.0008	0.0021	0.0006	0.00025	0.097	33	201.7	199	5.5	0.00005	0.0024	0.0025	0.0025	0.0025	0.0029	0.005	0.116	0.00006
Domestic #1 flow		49.5	gpm																								
Domestic #2 flow		500	gpm																								
Flow Weighted Average		0.0025	0.0048	0.0001	0.0012	0.0001	0.0016	0.0871	0.0032	0.0009	0.0027	0.0007	0.0004	0.1432	33.1158	227	325	6.34	0.0001	0.0024	0.0025	0.0025	0.0025	0.0031	0.0050	0.1097	0.0001